Week of 9th Feb

Monday, 9 February 2026 10:30 AM

→ Molecular partition functions

Some choice of p & q at a certain energy

Sum of microstates individually for non-interacting particles

$$q = \sum_{\nu} e^{-\beta \nu E}$$

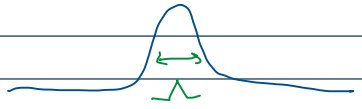
→ Third law of Thermodynamics

$T \to 0, \ \beta \to \infty$

Momenta can be split w/ position for integration.

$$Q = \frac{q^N}{N!} = \frac{V^N}{N! \Lambda^{3N}}$$

Canonical vs. Micro-canonical ??



→ Energy levels are decoupled

$$H = H_{trans} + H_{elec} + H_{rvib}$$

This can be assumed because the scale of things is very different. → Energy levels are ~ 10 eV while translation can occur & smaller eV's

Λ_i → de-Broglie wavelength

$$\Lambda_i = \left(\frac{h^2}{2\pi m_i kT} \right)^{1/2}$$

→ Jacobian does not apply to phase space transformations

$$p = kT \left. \frac{\partial \ln Q}{\partial V} \right|_{N,T}$$

$$Q = \sum_{\nu} e^{-\beta E_\nu} \qquad P_\nu = \frac{e^{-\beta E_\nu}}{Q}$$

$$U = \langle E_\nu \rangle = \sum_{\nu} P_\nu E_\nu = \frac{\sum_{\nu} E_\nu e^{-\beta E_\nu}}{} = \left. \frac{-\partial \ln Q}{\partial \beta} \right|$$

$$U = \langle E_V \rangle = \sum_V p_V E_V = \frac{\sum_V E_V e^{-\beta E_V}}{Q} = \left. \frac{-\partial \ln Q}{\partial \beta} \right|_{N,V}$$

C_V tells us the size of energy fluctuations of a system $\quad \langle (\delta E_V)^2 \rangle = kT^2 C_V$

At critical points the fluctuations in energy become meaningful

↳ $C_V \to \infty$ $\quad dV = C_V dT$

The states are being sampled extensively

where V population of states

temp. is used to vary the

→ Maxwell's demon

Schrodinger's Book → What is life?

Why are atoms so small?

→ Equipartition Theorem

Rotation Partition Function

$$q_{rot} = \frac{I}{\theta \sigma_{rot}}$$ (Classical limit for a diatomic)

Rotational temperature → ?

Rotational temperature → ??

Moment of inertia

↳ Finding degrees of freedom ; Symmetry no.

Potential Energy Surface

→ Multidimensional surface

→ @ const T, Boltzmann factor ($-\beta E$) indicates likely configurations

$\frac{kT}{h}$ → partition function for translation for the activated complex